

Units-of-output depreciation is the sum of the years of useful life (as of the beginning of which the numerator is divided by the estimated units of lifetime output. The unit depreciation cost is multiplied by the actual units of output each year to compute the annual depreciation expense.

DEMONSTRATION PROBLEM FOR YOUR REVIEW

On April 1, 1993, Argo Industries purchased new equipment at a cost of \$325,000. Useful life of this equipment was estimated at 5 years, with a residual value of \$25,000. For income tax purposes, however, this equipment is classified as "3-year property."

- 5 Compute the annual depreciation expense for each year until this equipment becomes fully depreciated under each depreciation method listed below. (Because you will record depreciation for only a fraction of a year in 1993, depreciation will extend through 1998 in all methods except MACRS.) Show supporting computations.
- a Straight-line, with depreciation for fractional years rounded to the nearest whole month. *(na)*
 - ☒ b 200%-declining-balance, with the half-year convention. Limit depreciation in 1998 to an amount which reduces the undepreciated cost to the estimated residual value.
 - c Sum-of-the-years'-digits, with the half-year convention.
 - d MACRS accelerated rates for "3-year property."

SOLUTION TO DEMONSTRATION PROBLEM

Year	Method of Depreciation			
	a Straight-Line	b 200%-Declining-Balance	c Sum-of-the-Years'-Digits	d MACRS
1993	\$ 45,000	\$ 65,000	\$ 50,000	\$108,322.50
1994	60,000	104,000	90,000	144,462.50
1995	60,000	62,400	70,000	48,132.50
1996	60,000	37,440	50,000	24,082.50
1997	60,000	22,464	30,000	-0-
1998	15,000	8,696	10,000	-0-
Totals	<u>\$300,000</u>	<u>\$300,000</u>	<u>\$300,000</u>	<u>\$325,000.00</u>

Supporting computations:

a

$$1993: (\$325,000 - \$25,000) \times \frac{1}{3} \times \frac{1}{2} = \$45,000$$

$$1994-1997: \$300,000 \times \frac{1}{3} = \$60,000$$

$$1998: \$300,000 \times \frac{1}{3} \times \frac{1}{2} = \$15,000$$

b

	Under-preciated Cost	Rate	Depreciation Expense
1993:	$\$325,000 \times 40\% \times \frac{1}{2} =$		\$65,000
1994:	$260,000 \times 40\% =$		104,000
1995:	$156,000 \times 40\% =$		62,400
1996:	$93,600 \times 40\% =$		37,440
1997:	$56,160 \times 40\% =$		22,464
1998:	$\boxed{33,696 - \$25,000} =$		8,696

d

1993: $\$300,000 \times \frac{1}{3} \times \frac{1}{2} =$	<u>\$50,000</u>	1993: $\$325,000 \times 33.33\% =$	\$108,322.50
1994: $\$300,000 \times \frac{1}{3} \times \frac{1}{2} =$	<u>\$50,000</u>	1994: $325,000 \times 44.45\% =$	144,462.50
$+ \$300,000 \times \frac{1}{3} \times \frac{1}{2} =$	<u>40,000</u>	1995: $325,000 \times 14.81\% =$	48,132.50
1995: $\$300,000 \times \frac{1}{3} \times \frac{1}{2} =$	<u>\$40,000</u>	1996: $325,000 \times 7.41\% =$	24,082.50
$+ \$300,000 \times \frac{1}{3} \times \frac{1}{2} =$	<u>30,000</u>		
1996: $\$300,000 \times \frac{1}{3} \times \frac{1}{2} =$	<u>\$30,000</u>		
$+ \$300,000 \times \frac{1}{3} \times \frac{1}{2} =$	<u>20,000</u>		
1997: $\$300,000 \times \frac{1}{3} \times \frac{1}{2} =$	<u>\$20,000</u>		
$+ \$300,000 \times \frac{1}{3} \times \frac{1}{2} =$	<u>10,000</u>		
1998: $\$300,000 \times \frac{1}{3} \times \frac{1}{2} =$	<u>\$10,000</u>		

SELF-TEST QUESTIONS

The answers to these questions appear on page 506.

- In which of the following situations should the named company **not** record any depreciation expense on the asset described?
 - Commuter Airline is required by law to maintain its aircraft in "as good as new" condition.

and transmission, at a cost of \$4,000. By making this expenditure it was believed that the useful life of the truck would be extended from the original estimate of 4 years to a total of 6 years. -

EXERCISE 10-4
Units-of-Output
Method

During the current year, Airport Auto Rentals purchased 60 new automobiles at a cost of \$13,000 per car. The cars will be sold to a wholesaler at an estimated \$4,000 each as soon as they have been driven 50,000 miles. Airport Auto Rentals computes depreciation expense on its automobiles by the units-of-output method, based upon mileage.

INSTRUCTIONS

- Compute the amount of depreciation to be recognized for each mile that a rental automobile is driven.
- Assuming that the 60 rental cars are driven a total of 1,650,000 miles during the current year, compute the total amount of depreciation expense that Airport Auto Rentals should recognize on this fleet of cars for the year.

EXERCISE 10-5
Double-
Declining-
Balance
Method

Machinery with an estimated useful life of 5 years was acquired by VPI Industries at a cost of \$55,000 at the beginning of the year. The estimated residual value of the machinery is \$6,000. Compute the annual depreciation on this machinery for each of the 5 years using the double-declining-balance method. Compute one full year's depreciation in each year. Limit the depreciation recognized in the fifth year to an amount that will cause the book value of the machinery to equal the estimated \$6,000 residual value at year-end.

EXERCISE 10-6
Sum-of-the-
Years'-Digits
Method

On January 2, Bartel Company acquired a machine at a cost of \$14,000. The machine is expected to have a useful life of 5 years with a residual value of \$2,000. You are to compute the annual depreciation on the machine in each of the 5 years of its useful life using the sum-of-the-years'-digits method. (One full year's depreciation will be taken each year.)

EXERCISE 10-7
Three Depreci-
ation Methods

Delta Company acquired new equipment with an estimated useful life of 5 years. Cost of the equipment was \$50,000 and the residual salvage value was estimated to be \$5,000. Compute the annual depreciation expense for each of the first 2 years under each of the following methods of depreciation. (Compute on full year's depreciation in each year.)

- Straight-line
- Sum-of-the-years'-digits
- Double-declining-balance

EXERCISE 10-8
Depreciation
for Fractional
Years

On November 2, Glass Recycling Company purchased special-purpose equipment at a cost of \$600,000. Useful life of the equipment was estimated to be 5 years and the residual value \$90,000. Compute the depreciation expense to be recognized in each calendar year during the life of the equipment under each of the following methods:

- Straight-line (round computations for a partial year to the nearest full month).
- Straight-line (use the half-year convention).
- The Modified Accelerated Cost Recovery System (MACRS). Use the table on page 481, and assume that the equipment qualifies as "3-year property."